



**3D Force Button Sensor
ROS2
(Beta v1.0)**

Installation and Operation Manual

Document #: 3DFBSROS2_B1.0_MAN_DEC24

December, 2024

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1 Introduction

The 3D Force Button Sensor Development Kit (Beta v1.0) is a system of (up to) five 3D Force Button Sensors per adapter, and a DEV001 development board. Each 3D Force Button Sensor can measure 3D force. The DEV001 development board supplies power for, and coordinates the simultaneous data acquisition from, up to five 3D Force Button Sensors. The DEV001 development board is shipped with visualisation software and C++ libraries for Windows and Linux environments and a ROS node for developing software control algorithms using the sensor signals.

Two 3D Force Button Sensors are shown in Figure 1.1, connected to a DEV001 development Board with USB-C cable.



Figure 1.1 – Two 3D Force Button Sensors connected to a DEV001 development board and USB-C cable.

This document is an installation and operation manual for the ROS2 Node which was provided on the Contactile USB flash drive that was shipped with the 3D Force Button Development Kit and DEV001 development board.

2 Safety

2.1 General

The customer should verify that the maximum loads and moments expected during operation fall within the sensing range of the sensor as outside this range, sensor reading accuracy is not guaranteed (refer to Document #3DFBS_Datasheet_RevX.X). Particular attention should be paid to dynamic loads caused by robot acceleration and deceleration if the sensors are mounted on robotic equipment. These forces can be many multiples of the value of static forces in high acceleration or deceleration situations.

2.2 Explanation of warnings

The warnings included here are specific to the product(s) covered by this manual. It is expected that the user heed all warnings from the manufacturers of other components used in the installation.



Danger indicates that a situation could result in potentially serious injury or damage to equipment.



Caution indicates that a situation could result in damage to the product and/or the other system components.

2.3 Precautions



DANGER: Do not attempt to disassemble the sensor. This could damage the sensor and will invalidate the calibration.



DANGER: Do not attempt to drill, tap, machine, or otherwise modify the sensor casing. This could damage the sensor and will void invalidated the calibration.



DANGER: Do not use the sensor on abrasive surfaces or surfaces with sharp points/edges. This could damage the silicone surface of the sensor.



CAUTION: Sensors may exhibit a small offset in readings when exposed to intense light sources.



CAUTION: Exceptionally strong and changing electromagnetic fields, such as those produced by magnetic resonance imaging (MRI) machines, constitute a possible source of interference with the operation of the sensor and Controller.



CAUTION: Temperature variations can cause drift in sensor readings. Some temperature compensation is performed. However, bias removal in software prior to operation is necessary, and it is recommended that biasing is performed each time the sensor is known to be unloaded.

3 Getting started

This section contains instructions for setting up and using 3D Force Button Sensor C++ Library for WINDOWS (Beta v1.0). It is recommended that first time users first read the preceding Safety section, then read through this section to get more familiar with the system.

3.1 Hardware installation

The ROS2 node is used with the 3D Force Button Sensors and DEV001 board. The DEV001 board should first be connected to the sensors and then to a PC via the USB-C port on the DEV001 board before you can use the visualisation software. For more information about connecting the sensors, refer to Document #DEV001_Datasheet_Rev1.1.

3.2 Minimum requirements

The ROS2 node has been tested on a Lenovo ThinkPad with the following specifications

- CPU: Intel Core i5-1240P
- RAM: 16 GB RAM
- Operating System: Ubuntu 20.04.2.0 LTS
- ROS2 installation: foxy

3.3 ROS2 node location

The ROS2 node is provided on the Contactile USB flash drive that was shipped with the development kit in the folder named SOFTWARE/ROS2. The ROS2 node is the *ros2_contactile_sensors* subfolder. The files in the *ROS2/ros2_contactile_sensors* folder are summarised in Table 3.1.

Table 3.1 – Files in ROS2/ros2_contactile_sensors

Sub Folder	Sub Sub Folder	File Name	File Description
.	.	README.md	Readme file
buttonsens or_ros2_v1	.	CMakeLists.txt	Directives and instructions describing the project's source files and targets
buttonsens or_ros2_v1	.	package.xml	The package manifest
buttonsens or_ros2_v1	src	buttonsensor_ros2_node.cpp buttonsensor_ros2_node.hpp	The cpp and header file for the node
buttonsens or_ros2_v1	lib	libPTSDK.a PTSDKListener.h PTSDKSensor.h PTSDKPillar.h PTSDKConstants.h	The C++ library and associated header files
buttonsens or_ros2_v1	launch	buttonsensor.launch	A launch file
sensor_interfaces	.	CMakeLists.txt	Directives and instructions describing the project's source files and targets

Sub Folder	Sub Sub Folder	File Name	File Description
sensor_interfaces	.	package.xml	The package manifest
sensor_interfaces	msg	ButtonSensorState.msg	Message definitions of the ButtonSensorState message
sensor_interfaces	srv	BiasRequest.srv	Service definitions of the BiasRequest service

4 Package summary

This package implements a driver for the 3D Force Button Sensor Development Kit (Beta v1.0).

Maintainer status: maintained

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License: LGPLv3

The `buttonsensor_ros2_v1` package provides a ROS2 interface that publishes data from ten 3D Force Button Sensors connected to a Controller that is connected over USB. The package allows the following:

1. Connecting to the controller, sampling the sensors at 100, 500 or 1000 Hz and publishing the data
2. Biasing the sensors
3. Starting and stopping slip detection on the Controller

5 Installing the package

To use this package you should have [ROS2 up and running](#). To install the package follow the steps below.

1. Copy the `ros2_contactile_sensors` folder into the `src` folder of your ROS2 workspace.
2. Navigate to the root of your ROS2 workspace.
3. Build `sensor_interfaces` package using command below:
`colcon build --packages-select sensor_interfaces`
4. Build `papillaryarray_ros2_v2` package using command below:
`colcon build --packages-select buttonsensor_ros2_v1`

6 Starting the node

The `buttonsensor.launch` launch file is provided to configure the COM port connection, get data from the sensors and publish the data.

The node can be started using:

```
ros2 launch buttonsensor_ros2_v1 buttonsensor.launch.py
```

NB: This will only work if you have completed the following steps:

1. Properly source your ROS2 installation and ROS2 Workspace
2. Allowed permission to access COM ports.

7 buttonsensor_ros2_node

ROS2-Node for connecting to the controller and publishing sensor data.

7.1 Parameters

Table 7.1 – Parameters of the buttonsensor_ros2_v1 node

Name	Type	Description
hub_id	int	Identifier for the Controller
n_sensors	int	Number of sensors connected to the Controller. Should be 10.
com_port	string	Name of the COM port. Usually /dev/ttyACM0.
baud_rate	int	Baud rate for the serial connection to the Controller. Should be 9600.
parity	int	Parity of the serial connection to the Controller. Should be 0.
byte_size	int	Size of a byte. Should be 8.
log_file_rate	int	Log file rate of the Controller. Options: 100, 500 or 1000 (Hz)

NB: Even though, only five 3DFB Sensors can be connected to the DEV001 development board, n_sensors must be 10 for backward compatibility with the older analogue 3D Force Button Sensors.

7.2 Subscribed topics

/hub_0/sensor_0 (See SensorState message)

- Publish sensor data from Sensor 0 connected to the DEV001 development board (PORT 0)

/hub_0/sensor_1 (See SensorState message)

- Publish sensor data from Sensor 1 connected to the DEV001 development board (PORT 0)

...

/hub_0/sensor_4 (See SensorState message)

- Publish sensor data from Sensor 4 connected to the DEV001 development board (PORT 4)

NB: The following topics also exist, but these will contain data equal to zero – these are required for backward compatibility with the older analogue 3D Button Force Sensor:

/hub_1/sensor_5 (See SensorState message)

/hub_1/sensor_6 (See SensorState message)

...

/hub_1/sensor_9 (See SensorState message)

7.3 Messages

7.3.1 ButtonSensorState message

Table 7.2 – Variables in the ButtonSensorState message

Parameter	Description
int64 tus	The time in μ s on the Controller of this sample
float32 gfx	The global X-force of the sensor
float32 gfy	The global Y-force of the sensor
float32 gfz	The global Z-force of the sensor

7.4 Services

Table 7.3 – Services of the buttonsensor_ros2_v1 node

Name	Description
SendBiasRequest	▪ Biasing refers to removing any offset in the sensor readings when the sensors are unloaded. ▪ It is recommended that the user performs a bias each time the sensors are known to be unloaded. ▪ Ensure that the sensor has been unloaded for at least one second before performing a bias to ensure that the bias calculation does not include hysteresis effects. A bias operation can take up to two seconds. Ensure that the sensor remains unloaded throughout this time.

7.5 Log file

7.5.1 Overview

In the `buttonsensor_ros2_v1/src/buttonsensor_ros2_node.cpp` file (line 3) a `PTSDKListener` object (`listener_`) is initialised with a boolean argument that specifies whether to log data to a .csv file or not. The argument can be changed to `true` (to enable logging) or `false` (to disable logging), remembering that the node should be remade by a call to the `catkin_make` utility.

7.5.2 Log file location

If logging is enabled, then a .csv log file will be generated and saved in the location `/home/.ros/Logs`.

NB: The `.ros` folder is a hidden location – make sure that you can view hidden files.

7.5.3 Log file name

The name of the log file that is generated is `LOG_YYYY_MM_DD_hh_mm_ss.csv` where:

- YYYY is the four digit year,
- MM is the two digit month,
- DD is the two digit day,
- hh is the two digit hour,
- mm is the two digit minute and
- ss is the two digit second,

from the system clock at the time that the log file was created.

7.5.4 Log file format

The log file is saved as comma-separated values (CSV) in ASCII text format. The order of the values and a description is shown in Table 7.1.

Table 7.1 – Data in log file

Data Order	Data Name	Data Description	
1	T_us	Timestamp in μ s	Timestamps
2	S0_G_FX	Sensor 0, X-axis force in Newtons	Sensor 0, forces
3	S0_G_FY	Sensor 0, Y-axis force in Newtons	
4	S0_G_FZ	Sensor 0, Z-axis force in Newtons	
⋮			
29	S9_G_FX	Sensor 9, X-axis force in Newtons	Sensor 9, forces
30	S9_G_FY	Sensor 9, Y-axis force in Newtons	
31	S9_G_FZ	Sensor 9, Z-axis force in Newtons	

S0, refers to the sensor connected to PORT 0, S1 refers to the sensor connected to PORT 1, and so on, and S4 refers to the sensor connected to PORT 4 of the DEV0001 development.

NB: S5 to S9 sensors will have all data equal to zero – these are necessary for backward compatibility with the older analogue 3D Force Button Sensor.

If a sensor is not connected, the log file will contain values of 0.0 for FX, FZ and FZ in the corresponding data columns.